Exercises week 2 – Answers

8CM10 - Introduction to Quantum Physics and Structure of Matter

Compton Scattering

1. A photon with 10 keV energy collides with a free electron at rest and is scattered through an angle of 60 deg. Compute:
    a. The change in energy and frequency of the photon

$$\lambda' - \lambda \frac{h}{m_e c}(1 - \cos\theta)$$

With $\theta = 60° = \frac{\pi}{3}$. Since $\frac{h}{m_e c} = 2.426 \cdot 10^{-3}$ and $\cos\frac{\pi}{3} = \frac{1}{2}$
The incoming photon has an energy $E = 10$ keV, and hence the wavelength:
$\lambda = \frac{hc}{E} = \frac{1240}{10^4} = 124 \cdot 10^{-3}$ nm
The frequency corresponding with this is $f = \frac{c}{\lambda} = \frac{2.998 \cdot 10^8}{124 \cdot 10^{-3}} = 2.4177 \cdot 10^{18}$ Hz
For the scattered photon, we obtain by plugging λ in the first equation:

$$\lambda' = 125.213 \cdot 10^{-3} \text{ nm}$$

Using the same formulae for energy and frequency but with λ' instead of λ:
$E' = 9.903$ keV and $f' = 2.3943 \cdot 10^{18}$ Hz.

    b. The kinetic energy, momentum and direction of the electron

We start by setting up the momentum balance for this system. The photon is labeled with a subscript p, the electron with a subscript e, and and final momenta for each of the components with an apostrophe.

$$p_{x,init} = p_{x,fin} \Rightarrow p_p = p_p' \cos\theta + p_e' \cos\phi$$
$$p_{y,init} = p_{y,fin} \Rightarrow 0 = p_p' \sin\theta - p_e' \sin\phi$$

We find the scattering angle ϕ the following way:

$$p_e' \cos\phi = p_p - p_p' \cos\theta$$
$$p_e' \sin\phi = p_p' \sin\theta$$

Which leads to after division of the equations:

$$\tan\phi = \frac{p_p' \sin\theta}{p_p - p_p' \cos\theta} \Rightarrow \phi = \arctan\left(\frac{p_p' \sin\theta}{p_p - p_p' \cos\theta}\right) = \arctan\left(\frac{E_p' \sin\theta}{E_p - E_p' \cos\theta}\right) = 59.5164°$$

Where in the last step we multiplied the fraction with c on both sides used that for photons combined with $E = pc$ for photons.

The momentum of the electron p_e' is found by combining the energy-momentum relationship with relativistic kinetic energy. The difference in energy between incoming and scattered photon is the kinetic energy K_e of the electron.
Hence $K_e = 10 - 9.903\ \text{keV} = 97\ \text{eV}$.

Working out the relevant formulae yields:

Relativistic kinetic energy: $$E_e^2 = \left(K_e + E_{0,e}\right)^2 = K_e^2 + 2KE_{0,e} + E_{0,e}$$

Energy momentum: $$E_e^2 = (p_e')^2c^2 + E_{0,e}^2 \Rightarrow p_e' = \frac{\sqrt{E_e^2 - E_{0,e}^2}}{c}$$

Combining: $$p_e' = \frac{\sqrt{K_e^2 + 2KE_{0,e}}}{c} = \sqrt{97^2 + 2 \cdot 97 \cdot 511 \cdot 10^3}\ \text{eV/c}$$
$$= 9.95708\ \text{keV/c} = 9.95708 \cdot 10^3 \cdot \frac{1.602 \cdot 10^{-19}}{2.998 \cdot 10^8}\ \text{kg} \cdot \text{m/s}$$
$$= 5.3206 \cdot 10^{-24}\ \text{kg} \cdot \text{m/s}$$

This calculation has used relativistic mechanics. Using classical mechanics would give a momentum of $5.3207 \cdot 10^{-24}\ \text{kg} \cdot \text{m/s}$, so in this particular case classical mechanics could have been used.

2. X-ray photons of wavelength 0.02480 nm are incident on a target and the Compton-scattered photons are observed at 90.0◦.
    a. What is the wavelength of the scattered photons?
    b. What is the momentum of the incident photons? Of the scattered photons?
    c. What is the kinetic energy of the scattered electrons?
    d. What is the momentum (magnitude and direction) of the scattered electrons?

(a) $\lambda' = \lambda + (h/m_e c)(1-\cos\theta)$

$= 0.02480\ \text{nm} + (0.002426\ \text{nm})(1-\cos 90°) = 0.02723\ \text{nm}$

(b) The momenta of the incident and scattered photons are

$$p = \frac{h}{\lambda} = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{eV}\cdot\text{nm}}{0.02480\ \text{nm}} = 4.999\times10^4\ \text{eV}/c \quad (x\ \text{direction})$$

$$p' = \frac{h}{\lambda} = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{eV}\cdot\text{nm}}{0.02723\ \text{nm}} = 4.553\times10^4\ \text{eV}/c \quad (y\ \text{direction})$$

(c) $K_e = E - E' = cp - cp' = 4.999\times10^4\ \text{eV} - 4.553\times10^4\ \text{eV} = 4.46\times10^3\ \text{eV}$

(d) Because momentum must be conserved, the x component of the electron's momentum must equal p, and the y component must equal $-p'$:

$$p_{ex} = p = 4.999\times10^4\ \text{eV}/c \qquad \text{and} \qquad p_{ey} = -p' = -4.553\times10^4\ \text{eV}/c$$

$$p_e = \sqrt{p_{ex}^2 + p_{ey}^2} = \sqrt{(4.999\times10^4\ \text{eV}/c)^2 + (4.553\times10^4\ \text{eV}/c)^2} = 6.762\times10^4\ \text{eV}/c$$

in the direction given by $\theta = \tan^{-1}\dfrac{p_{ey}}{p_{ex}} = \tan^{-1}\dfrac{-4.553\times10^4\ \text{eV}/c}{4.999\times10^4\ \text{eV}/c} = -42.3°$

3. Gamma rays of energy 0.662MeV are Compton scattered.
   a. What is the energy of the scattered photon observed at a scattering angle of 60.0°?
   b. What is the kinetic energy of the scattered electrons?

(a) $$\frac{1}{E'} = \frac{1}{E} + \frac{1-\cos\theta}{m_e c^2} = \frac{1}{0.662\ \text{MeV}} + \frac{1-\cos 60°}{0.511\ \text{MeV}} = 2.489\ \text{MeV}^{-1}$$

so $E' = 1/2.489\ \text{MeV}^{-1} = 0.402\ \text{MeV}$.

(b) $K_e = E - E' = 0.662\ \text{MeV} - 0.402\ \text{MeV} = 0.260\ \text{MeV}$

De Broglie Wavelengths

4. By doing a nuclear diffraction experiment, you measure the de Broglie wavelength of a proton to be 9.16 fm.
    a. What is the speed of the proton?
    b. Through what potential difference must it be accelerated to achieve that speed?

(a) $p = \frac{h}{\lambda} = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{MeV}\cdot\text{fm}}{9.16\ \text{fm}} = 135.4\ \text{MeV}/c$

From $p = mv/\sqrt{1-v^2/c^2} = (1/c)mc^2(v/c)/\sqrt{1-v^2/c^2}$ we solve for v:

$$v = \frac{c}{\sqrt{1+(mc^2/pc)^2}} = \frac{c}{\sqrt{1+[(938.3\ \text{MeV})/(135.4\ \text{MeV})]^2}} = 0.143c$$

(b) $$K = E - E_0 = \sqrt{(pc)^2 + (mc^2)^2} - mc^2$$
$$= \sqrt{(135.4\ \text{MeV})^2 + (938.3\ \text{MeV})^2} - 938.3\ \text{MeV} = 9.72\ \text{MeV}$$

This gain in kinetic energy requires a loss in potential energy of $\Delta U = -9.72$ MeV and thus a potential difference of $\Delta V = \Delta U/q = -9.72\ \text{MeV}/e = -9.72$ MV .

5. Find the potential difference through which electrons must be accelerated (as in an electron microscope, for example) if we wish to resolve:
    a. a virus of diameter 12 nm
    b. an atom of diameter 0.12 nm
    c. a proton of diameter 1.2 fm.

(a) For $\lambda = 12$ nm,

$$p = \frac{h}{\lambda} = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{eV}\cdot\text{nm}}{12\ \text{nm}} = 100\ \text{eV}/c$$

$$K = \frac{p^2}{2m} = \frac{p^2c^2}{2mc^2} = \frac{(100\ \text{eV})^2}{2(511{,}000\ \text{eV})} = 0.010\ \text{eV}$$

To increase the kinetic energy by $\Delta K = 0.010$ eV, its potential energy must decrease by $\Delta U = -\Delta K$, where $\Delta U = q\,\Delta V$:

$$\Delta V = \frac{\Delta U}{q} = \frac{-\Delta K}{q} = \frac{-0.010\ \text{eV}}{-e} = +0.010\ \text{V}$$

(b) For $\lambda = 0.12$ nm,

$$p = \frac{h}{\lambda} = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{eV}\cdot\text{nm}}{0.12\ \text{nm}} = 1.0\times10^4\ \text{eV}/c$$

$$K = \frac{p^2}{2m} = \frac{p^2c^2}{2mc^2} = \frac{(1.0\times10^4\ \text{eV})^2}{2(511{,}000\ \text{eV})} = 100\ \text{eV}$$

$$\Delta V = -\Delta K / q = +100\ \text{V}$$

(c) For $\lambda = 1.2$ fm,

$$p = \frac{1}{c}\frac{hc}{\lambda} = \frac{1}{c}\frac{1240\ \text{MeV}\cdot\text{fm}}{1.2\ \text{fm}} = 1000\ \text{MeV}/c$$

Here $pc \gg mc^2$, so we can use the extreme relativistic approximation:

$$K \cong E \cong pc = 1000\ \text{MeV} = 1.0\times10^9\ \text{eV}$$

$$\Delta V = -\Delta K / q = +1.0\times10^9\ \text{V}$$

Although it is possible to accelerate electrons to such high energies, it is not done by a single acceleration through such a large potential difference.

Uncertainty Principle

6. Suppose a traveling wave has a speed v (where $v = \lambda f$). Instead of measuring waves over a distance Δx, we stay in one place and count the number of wave crests that pass in a time Δt. Show that $\Delta f \Delta t \sim \epsilon$ is equivalent to $\Delta x \Delta\lambda \sim \epsilon\lambda^2$ for this case (see 4.3 in Krane).

The number of wave crests passing the observation point in time Δt is $N = f\,\Delta t$. The uncertainty in this number is

$$\Delta N = \Delta f\,\Delta t$$

With $f = v/\lambda$, we take differentials and obtain

$$df = -\frac{v}{\lambda^2}d\lambda \qquad \text{or} \qquad \Delta f = \frac{v}{\lambda^2}\Delta\lambda$$

Using $\Delta t = \Delta x / v$ (where Δx is the distance traveled by the wave in the time Δt),

$$\Delta N = \Delta f\,\Delta t = \left(\frac{v}{\lambda^2}\Delta\lambda\right)\left(\frac{\Delta x}{v}\right) = \frac{\Delta x\,\Delta\lambda}{\lambda^2}$$

If the uncertainty in counting is a fraction ε of one wave ($\Delta N \sim \varepsilon$), then these results are consistent with $\Delta x\,\Delta\lambda \sim \varepsilon\lambda^2$ and $\Delta f\,\Delta t \sim \varepsilon$.

7. The lifetime of an excited state is 10^{-8}s. What is the minimum uncertainty in
   a. The energy
   b. The frequency

If the lifetime of an excited state is 10^{-8} s, then we have only 10^{-8} s to measure the energy. That means that the uncertainty relation $\Delta E\Delta t \geq \frac{\hbar}{2}$ can be used with $\Delta t = 10^{-8}$ in order to calculate the minimum uncertainty.

a) Substitution in $\Delta E\Delta t \geq \frac{\hbar}{2}$ gives $\Delta E \geq 5.25 \cdot 10^{-27}$J.

b) From $E = hf$ it follows that $\Delta E = h\Delta f$ so that:

$$\Delta f \geq \frac{\hbar}{2h\Delta t} = \frac{h}{2\pi \cdot 2h\Delta t} = \frac{1}{4\pi \cdot 10^{-8}} \Rightarrow \Delta f \geq \frac{10^8}{4\pi} = 7.958 \cdot 10^6 \text{ Hz}$$

8. The speed of an electron is measured to within an uncertainty of $2.0 \cdot 10^4$ m/s. What is the size of the smallest region of space in which the electron can be confined?

With $\Delta v = 2.0 \times 10^4$ m/s,

$$\Delta x \sim \frac{\hbar}{\Delta p} = \frac{\hbar}{m\,\Delta v} = \frac{1.05\times10^{-34}\text{ J}\cdot\text{s}}{(9.11\times10^{-31}\text{ kg})(2.0\times10^4\text{ m/s})} = 5.8\times10^{-9}\text{ m} = 5.8\text{ nm}$$